Jonathan Ullman

Curriculum Vitae

Associate Professor
Khoury College of Computer Sciences
Northeastern University

j.ullman@northeastern.edu
https://jonathan-ullman.github.io
November 2026

Education

Doctor of Philosophy in Computer Science 08/2009–06/2013
Harvard University School of Engineering and Applied Sciences
Thesis Title: *Privacy and the Complexity of Simple Queries*
Thesis Advisor: Salil Vadhan

Bachelor of Science in Engineering (Computer Science) 08/2004–05/2008
Princeton University
Graduated *magna cum laude*

Employment History

Associate Professor 06/2021–Current
Khoury College of Computer Sciences
Northeastern University

Assistant Professor 08/2015–06/2021
Khoury College of Computer Sciences
Northeastern University

Junior Fellow 07/2014–07/2015
Simons Society of Fellows
Host: Rocco Servedio, Columbia University

Postdoctoral Fellow 06/2013–06/2014
Center for Research on Computation and Society
Harvard University

Research Intern 08/2011–12/2011
Microsoft Research SVC
Host: Cynthia Dwork

Honors and Awards

Apple Research Award 02/2021
Project Title: *Auditing Differentially Private Machine Learning*

Ruth and Joel Spira Outstanding Teacher Award 09/2019
Awarded annually to a faculty member in the Khoury College of Computer Sciences

NSF CAREER Award 02/2018
Project Title: *A Stable Foundation for Trustworthy Data Analysis*

Google Faculty Research Award 02/2018
Project Title: *Distributed Differential Privacy Beyond Local Protocols*

Publications

Primary Conference and Journal Publications (Reverse Chronological Order)

My work is primarily published in computer science conferences, and this list includes all primary conference and journal publications. Author order is alphabetical unless otherwise noted.

[1] Anagha Gokul, Jason Hartline, Lunjia Hu, Jonathan Ullman, and Yifan Wu. "Truthful Calibration Measures for Sequential Prediction." *ACM-SIAM Symposium on Discrete Algorithms*. SODA 2027.

[2] Gautam Kamath, Mahbod Majid, Ankur Moitra, Argyris Mouzakis, and Jonathan Ullman. "Efficient and Optimal Learning of Discrete Distributions with Person-Level Privacy." *ACM-SIAM Symposium on Discrete Algorithms*. SODA 2027.

[3] Konstantina Bairaktari, Lunjia Hu, Huy Lê Nguyễn, and Jonathan Ullman. "Testable and Actionable Calibration for Full Swap Regret." *Annual Conference on Neural Information Processing Systems*. NeurIPS 2026. Oral Presentation (top 0.4% of submissions).

[4] Mahdi Haghifam, Adam Smith, and Jonathan Ullman. "The Sample Complexity of Membership Inference and Privacy Auditing." *IEEE Symposium on Foundations of Computer Science*. FOCS 2026.

[5] Aleksandar Nikolov, Haohua Tang, and Jonathan Ullman. "Online Matrix Factorization, Online Private Query Release, and Online Discrepancy Minimization." *ACM Symposium on Theory of Computing*. STOC 2026.

[6] John Abascal, Nicolás Berrios, Alina Oprea, Jonathan Ullman, Adam Smith, and Matthew Jagielski. "Black-Box Privacy Attacks on Shared Representations in Multitask Learning." *International Conference on Learning Representations*. ICLR 2026. (authors by contribution).

[7] Amrith Setlur, Pratiksha Thaker, and Jonathan Ullman. "Lower Bounds for Public-Private Learning under Distribution Shift." *International Conference on Artificial Intelligence and Statistics*. AISTATS 2026.

[8] Maryam Aliakbarpour, Konstantina Bairaktari, Adam Smith, Marika Swanberg, and Jonathan Ullman. "Privacy in Metalearning and Multitask Learning: Modeling and Separations." *International Conference on Artificial Intelligence and Statistics*. AISTATS 2025.

[9] Gautam Kamath, Argyris Mouzakis, Matthew Regehr, Vikrant Singhal, Thomas Steinke, and Jonathan Ullman. "A Bias-Accuracy-Privacy Trilemma for Statistical Estimation." *Journal of the American Statistical Association* (2025).

[10] Sushant Agarwal, Gautam Kamath, Mahbod Majid, Argyris Mouzakis, Rose Silver, and Jonathan Ullman. "Private Mean Estimation with Person-Level Differential Privacy." *ACM-SIAM Symposium on Discrete Algorithms*. SODA 2025.

[11] Mahdi Haghifam, Thomas Steinke, and Jonathan Ullman. "Private Geometric Median." *Annual Conference on Neural Information Processing Systems*. NeurIPS 2024.

[12] Liudas Panavas, Tarik Crnovrsanin, Jane Lydia Adams, Jonathan Ullman, Ali Sargavad, Melanie Tory, and Cody Dunne. "Investigating the Visual Utility of Differentially Private Scatterplots." *IEEE Transactions on Visualization and Computer Graphics* 30.8 (2024). (authors by contribution).

[13] Maryam Aliakbarpour, Konstantina Bairaktari, Gavin Brown, Adam Smith, Nathan Srebro, and Jonathan Ullman. "Metalearning with Very Few Samples Per Task." *Conference on Learning Theory*. COLT 2024.

[14] Naty Peter, Eliad Tsfadia, and Jonathan Ullman. "Smooth Lower Bounds for Differentially Private Algorithms via Padding-and-Permuting Fingerprinting Codes." *Conference on Learning Theory*. COLT 2024.

[15] Andrew Lowy, Jonathan Ullman, and Stephen Wright. "How to Make the Gradients Small Privately: Improved Rates for Differentially Private Non-Convex Optimization." *International Conference on Machine Learning*. ICML 2024.

[16] Jiawen Liu, Weihao Qu, Marco Gaboardi, Deepak Garg, and Jonathan Ullman. "Program Analysis for Adaptive Data Analysis." *ACM SIGPLAN Conference on Programming Language Design and Implementation*. PLDI 2024. (authors by contribution).

[17] John Abascal, Stanley Wu, Alina Oprea, and Jonathan Ullman. "TMI! Finetuned Models Leak Private Information from their Pretraining Data." *Privacy Enhancing Technologies Symposium*. PETS 2024. (authors by contribution).

[18] Harsh Chaudhari, Giorgio Severi, Alina Oprea, and Jonathan Ullman. "Chameleon: Increasing Label-Only Membership Leakage with Adaptive Poisoning." *International Conference on Learning Representations*. ICLR 2024. (authors by contribution).

[19] Maryam Aliakbarpour, Rose Silver, Thomas Steinke, and Jonathan Ullman. "Differentially Private Medians and Interior Points for Non-Pathological Data." *Innovations in Theoretical Computer Science*. ITCS 2024.

[20] Konstantina Bairaktari, Paul Tsela Langton, Huy Lê Nguyễn, Niklas Smedemark-Margulies, and Jonathan Ullman. "Fair and Useful Cohort Selection." *Transactions on Machine Learning Research* (2023).

[21] Konstantina Bairaktari, Guy Blanc, Li-Yang Tan, Jonathan Ullman, and Lydia Zakynthinou. "Multitask Learning via Shared Features: Algorithms and Hardness." *Conference on Learning Theory*. COLT 2023.

[22] Hilal Asi, Jonathan Ullman, and Lydia Zakynthinou. "From Robustness to Privacy and Back." *International Conference on Machine Learning*. ICML 2023.

[23] Matthew Jagielski, Stanley Wu, Alina Oprea, Jonathan Ullman, and Roxana Geambasu. "How to Combine Membership-Inference Attacks on Multiple Updated Machine Learning Models." *Privacy Enhancing Technologies Symposium*. PETS 2023. (nonalphabetical author order).

[24] Harsh Chaudhari, John Abascal, Alina Oprea, Matthew Jagielski, Florian Tramèr, and Jonathan Ullman. "SNAP: Efficient Extraction of Private Properties with Poisoning." *IEEE Symposium on Security and Privacy*. IEEE S&P 2023. (authors by contribution).

[25] Gautam Kamath, Argyris Mouzakis, Vikrant Singhal, Thomas Steinke, and Jonathan Ullman. "A Private and Computationally Efficient Estimator for Unbounded Gaussians." *Conference on Learning Theory*. COLT 2022.

[26] Gavin Brown, Marco Gaboardi, Adam Smith, Jonathan Ullman, and Lydia Zakynthinou. "Covariance-Aware Private Mean Estimation Without Private Covariance Estimation." *Annual Conference on Neural Information Processing Systems*. NeurIPS 2021. Spotlight Presentation (top 2.9% of submissions).

[27] Terrance Liu, Giuseppe Vietri, Thomas Steinke, Jonathan Ullman, and Zhiwei Steven Wu. "Leveraging Public Data for Practical Private Query Release." *International Conference on Machine Learning*. ICML 2021. (nonalphabetical author order).

[28] Albert Cheu and Jonathan Ullman. "The Limits of Pan Privacy and Shuffle Privacy for Learning and Estimation." *ACM Symposium on Theory of Computing*. STOC 2021.

[29] Albert Cheu, Adam Smith, and Jonathan Ullman. "Manipulation Attacks in Local Differential Privacy." *IEEE Symposium on Security and Privacy*. IEEE S&P 2021.

[30] Matthew Jagielski, Jonathan Ullman, and Alina Oprea. "Auditing Differentially Private Machine Learning: How Private is Private SGD?" *Annual Conference on Neural Information Processing Systems*. NeurIPS 2020. (authors by contribution).

[31] Sourav Biswas, Yihe Dong, Gautam Kamath, and Jonathan Ullman. "CoinPress: Practical Private Mean and Covariance Estimation." *Annual Conference on Neural Information Processing Systems*. NeurIPS 2020.

[32] Clément L. Canonne, Gautam Kamath, Audra McMillan, Jonathan Ullman, and Lydia Zakynthinou. "Private Identity Testing for High Dimensional Distributions." *Annual Conference on Neural Information Processing Systems*. NeurIPS 2020. Spotlight Presentation (top 3.0% of submissions).

[33] Gautam Kamath, Vikrant Singhal, and Jonathan Ullman. "Private Mean Estimation of Heavy-Tailed Distributions." *Conference on Learning Theory*. COLT 2020.

[34] Raef Bassily, Albert Cheu, Shay Moran, Aleksandar Nikolov, Jonathan Ullman, and Zhiwei Steven Wu. "Private Query Release Assisted by Public Data." *International Conference on Machine Learning*. ICML 2020.

[35] Alexander Edmonds, Aleksandar Nikolov, and Jonathan Ullman. "The Power of Factorization Mechanisms in Local and Central Differential Privacy." *ACM Symposium on Theory of Computing*. STOC 2020.

[36] Huy Lê Nguyễn, Jonathan Ullman, and Lydia Zakynthinou. "Efficient Private Algorithms for Learning Halfspaces." *International Conference on Algorithmic Learning Theory*. ALT 2020.

[37] Gautam Kamath, Or Sheffet, Vikrant Singhal, and Jonathan Ullman. "Differentially Private Algorithms for Learning Mixtures of Well Separated Gaussians." *Annual Conference on Neural Information Processing Systems*. NeurIPS 2019.

[38] Adam Sealfon and Jonathan Ullman. "Efficiently Estimating Erdős-Rényi Graphs with Node Differential Privacy." *Annual Conference on Neural Information Processing Systems*. NeurIPS 2019.

[39] Jeffrey Champion, Abhi Shelat, and Jonathan Ullman. "Securely Sampling Biased Coins with Applications to Differential Privacy." *ACM Conference on Computer and Communications Security*. CCS 2019.

[40] Matthew Jagielski, Michael Kearns, Jieming Mao, Alina Oprea, Aaron Roth, Saeed Sharifi-Malvajerdi, and Jonathan Ullman. "Differentially Private Fair Learning." *International Conference on Machine Learning*. ICML 2019.

[41] Gautam Kamath, Jerry Li, Vikrant Singhal, and Jonathan Ullman. "Privately Learning High Dimensional Distributions." *Conference on Learning Theory*. COLT 2019.

[42] Clément Canonne, Gautam Kamath, Audra McMillan, Adam Smith, and Jonathan Ullman. "The Structure of Optimal Private Tests for Simple Hypotheses." *ACM Symposium on Theory of Computing*. STOC 2019.

[43] Albert Cheu, Adam Smith, Jonathan Ullman, David Zeber, and Maxim Zhilyaev. "Distributed Differential Privacy via Shuffling." *Annual International Conference on the Theory and Applications of Cryptographic Techniques*. EUROCRYPT 2019.

[44] Cynthia Dwork and Jonathan Ullman. "The Fienberg problem: How to allow human interactive data analysis in the age of differential privacy." *Journal of Privacy and Confidentiality* 8.1 (2018).

[45] Foto N. Afrati, Shantanu Sharma, Jonathan R. Ullman, and Jeffrey D. Ullman. "Computing Marginals Using MapReduce." *Journal of Computer and System Sciences* 94 (2018).

[46] Matthew Joseph, Aaron Roth, Jonathan Ullman, and Bo Waggoner. "Local Differential Privacy for Evolving Data." *Annual Conference on Neural Information Processing Systems*. NeurIPS 2018. Spotlight Presentation (top 3.5% of submissions).

[47] Kobbi Nissim, Adam Smith, Thomas Steinke, Uri Stemmer, and Jonathan Ullman. "The Limits of Post-Selection Generalization." *Annual Conference on Neural Information Processing Systems*. NeurIPS 2018.

[48] Albert Cheu, Ravi Sundaram, and Jonathan Ullman. "Skyline Identification in Multi-Armed Bandits." *IEEE International Symposium on Information Theory*. ISIT 2018.

[49] Lucas Kowalczyk, Tal Malkin, Jonathan Ullman, and Daniel Wichs. "Hardness of Non-Interactive Differential Privacy from One-Way Functions." *Annual International Cryptology Conference*. CRYPTO 2018.

[50] Mallesh M. Pai, Aaron Roth, and Jonathan Ullman. "An Antifolk Theorem for Large Repeated Games." *ACM Transactions on Economics and Computation (TEAC)* 5.2 (2017).

[51] Thomas Steinke and Jonathan Ullman. "Between Pure and Approximate Differential Privacy." *Journal of Privacy and Confidentiality* 7.2 (2017).

[52] Thomas Steinke and Jonathan Ullman. "Tight Lower Bounds for Differentially Private Selection." *IEEE Symposium on Foundations of Computer Science*. FOCS 2017.

[53] Piotr Indyk, Sepideh Mahabadi, Ronitt Rubinfeld, Jonathan Ullman, Ali Vakilian, and Anak Yodpinyanee. "Fractional Set Cover in the Streaming Model." *International Workshop on Approximation Algorithms for Combinatorial Optimization Problems*. APPROX 2017.

[54] Mitali Bafna and Jonathan Ullman. "The Price of Selection in Differential Privacy." *Conference on Learning Theory*. COLT 2017.

[55] Aaron Roth, Aleksandrs Slivkins, Jonathan Ullman, and Zhiwei Steven Wu. "Multidimensional Dynamic Pricing for Welfare Maximization." *ACM Conference on Economics and Computation*. EC 2017. Invited to a special issue of ACM Transactions on Economics and Computation.

[56] Mark Bun, Thomas Steinke, and Jonathan Ullman. "Make Up Your Mind: The Price of Online Queries in Differential Privacy." *ACM-SIAM Symposium on Discrete Algorithms*. SODA 2017.

[57] Ryan Rogers, Aaron Roth, Jonathan Ullman, and Salil Vadhan. "Privacy Odometers and Filters: Pay-as-You-Go Composition." *Annual Conference on Neural Information Processing Systems*. NeurIPS 2016.

[58] Lucas Kowalczyk, Tal Malkin, Jonathan Ullman, and Mark Zhandry. "Strong Hardness of Privacy from Weak Traitor Tracing." *Theory of Cryptography Conference*. TCC 2016-B.

[59] Edo Liberty, Michael Mitzenmacher, Justin Thaler, and Jonathan Ullman. "Space Lower Bounds for Itemset Frequency Sketches." *ACM Symposium on Principles of Database Systems*. PODS 2016.

[60] Raef Bassily, Kobbi Nissim, Adam Smith, Thomas Steinke, Uri Stemmer, and Jonathan Ullman. "Algorithmic Stability for Adaptive Data Analysis." *ACM Symposium on Theory of Computing*. STOC 2016. Invited to a special issue of SIAM Journal on Computing.

[61] Aaron Roth, Jonathan Ullman, and Zhiwei Steven Wu. "Watch and Learn: Optimizing from Revealed Preferences Feedback." *ACM Symposium on Theory of Computing*. STOC 2016.

[62] Cynthia Dwork, Adam Smith, Thomas Steinke, Jonathan Ullman, and Salil Vadhan. "Robust Traceability from Trace Amounts." *IEEE Symposium on Foundations of Computer Science*. FOCS 2015.

[63] Thomas Steinke and Jonathan Ullman. "Interactive Fingerprinting Codes and the Hardness of Preventing False Discovery." *Conference on Learning Theory*. COLT 2015.

[64] Jonathan Ullman. "Private Multiplicative Weights Beyond Linear Queries." *ACM Symposium on Principles of Database Systems*. PODS 2015.

[65] Pavel Hubáček, Moni Naor, and Jonathan Ullman. "When Can Limited Randomness Be Used in Repeated Games?" *International Symposium on Algorithmic Game Theory*. SAGT 2015. Invited to a special issue of Theory of Computing Systems.

[66] Ryan M. Rogers, Aaron Roth, Jonathan Ullman, and Zhiwei Steven Wu. "Inducing Approximately Optimal Flow Using Truthful Mediators." *ACM Conference on Economics and Computation*. EC 2015.

[67] Moritz Hardt and Jonathan Ullman. "Preventing False Discovery in Interactive Data Analysis Is Hard." *IEEE Symposium on Foundations of Computer Science*. FOCS 2014.

[68] Justin Hsu, Aaron Roth, Tim Roughgarden, and Jonathan Ullman. "Privately Solving Linear Programs." *International Colloquium on Automata, Languages, and Programming, Track A*. ICALP(A) 2014.

[69] Karthekeyan Chandrasekaran, Justin Thaler, Jonathan Ullman, and Andrew Wan. "Faster Private Release of Marginals on Small Databases." *Innovations in Theoretical Computer Science*. ITCS 2014.

[70] Michael Kearns, Mallesh M. Pai, Aaron Roth, and Jonathan Ullman. "Mechanism Design in Large Games: Incentives and Privacy." *Innovations in Theoretical Computer Science*. ITCS 2014.

[71] Mark Bun, Jonathan Ullman, and Salil Vadhan. "Fingerprinting Codes and the Price of Approximate Differential Privacy." *ACM Symposium on Theory of Computing*. STOC 2014. Invited to a special issue of SIAM Journal on Computing.

[72] Justin Hsu, Aaron Roth, and Jonathan Ullman. "Differential Privacy for the Analyst via Private Equilibrium Computation." *ACM Symposium on Theory of Computing*. STOC 2013.

[73] Jonathan Ullman. "Answering $n^{2+o(1)}$ Counting Queries with Differential Privacy is Hard." *ACM Symposium on Theory of Computing*. STOC 2013. Invited to a special issue of SIAM Journal on Computing.

[74] Justin Thaler, Jonathan Ullman, and Salil P. Vadhan. "Faster Algorithms for Privately Releasing Marginals." *International Colloquium on Automata, Languages, and Programming, Track A*. ICALP(A) 2012.

[75] Anupam Gupta, Aaron Roth, and Jonathan Ullman. "Iterative Constructions and Private Data Release." *Theory of Cryptography Conference*. TCC 2012.

[76] Anupam Gupta, Moritz Hardt, Aaron Roth, and Jonathan Ullman. "Privately releasing conjunctions and the statistical query barrier." *ACM Symposium on Theory of Computing*. STOC 2011.

[77] Jonathan Ullman and Salil P. Vadhan. "PCPs and the Hardness of Generating Private Synthetic Data." *Theory of Cryptography Conference*. TCC 2011. Invited to the Journal of Cryptology.

[78] Shiva Prasad Kasiviswanathan, Mark Rudelson, Adam Smith, and Jonathan Ullman. "The Price of Privately Releasing Contingency Tables and the Spectra of Random Matrices with Correlated Rows." *ACM Symposium on Theory of Computing*. STOC 2010.

[79] Scott Duke Kominers, Mike Ruberry, and Jonathan Ullman. "Course Allocation by Proxy Auction." *International Workshop on Internet and Network Economics*. WINE 2010.

Secondary Publications (Reverse Chronological Order)

Some of my work appears in journals as a secondary form of publication, after initially appearing in a computer science conference. This list includes all such publications.

[80] Raef Bassily, Kobbi Nissim, Adam Smith, Thomas Steinke, Uri Stemmer, and Jonathan Ullman. "Algorithmic stability for adaptive data analysis." *SIAM Journal on Computing* 50.3 (2021).

[81] Adam Sealfon and Jonathan Ullman. "Efficiently Estimating Erdős-Rényi Graphs with Node Differential Privacy." *Journal of Privacy and Confidentiality* 11.1 (2021).

[82] Albert Cheu, Adam Smith, and Jonathan Ullman. "Manipulation Attacks in Local Differential Privacy." *Journal of Privacy and Confidentiality* 11.1 (2021).

[83] Matthew Joseph, Aaron Roth, Jonathan Ullman, and Bo Waggoner. "Local Differential Privacy for Evolving Data." *Journal of Privacy and Confidentiality* 10.1 (2020).

[84] Jonathan Ullman and Salil Vadhan. "PCPs and the hardness of generating synthetic data." *Journal of Cryptology* (2020).

[85] Aaron Roth, Aleksandrs Slivkins, Jonathan Ullman, and Zhiwei Steven Wu. "Multidimensional Dynamic Pricing for Welfare Maximization." *ACM Transactions on Economics and Computation* 8.1 (2020).

[86] Mark Bun, Thomas Steinke, and Jonathan Ullman. "Make Up Your Mind: The Price of Online Queries in Differential Privacy." *Journal of Privacy and Confidentiality* 9.1 (2019).

[87] Mark Bun, Jonathan Ullman, and Salil Vadhan. "Fingerprinting Codes and the Price of Approximate Differential Privacy." *SIAM Journal on Computing* 47.5 (2018).

[88] Pavel Hubáček, Moni Naor, and Jonathan Ullman. "When can limited randomness be used in repeated games?" *Theory of Computing Systems* 59.4 (2016).

[89] Jonathan Ullman. "Answering $n^{2+o(1)}$ Counting Queries with Differential Privacy is Hard." *SIAM Journal on Computing* 45.2 (2016).

[90] Anupam Gupta, Moritz Hardt, Aaron Roth, and Jonathan Ullman. "Privately Releasing Conjunctions and the Statistical Query Barrier." *SIAM Journal on Computing* 42.4 (2013).

Surveys and Other Writing (Reverse Chronological Order)

[91] Clément Canonne, Gautam Kamath, Thomas Steinke, Jonathan Ullman, and Zhiwei Steven Wu. *DifferentialPrivacy.org*. 2020.

[92] Jonathan Ullman. "Technical Perspective: Building a Safety Net for Data Reuse." *Communications of the ACM* 60.4 (2017).

[93] Cynthia Dwork, Adam Smith, Thomas Steinke, and Jonathan Ullman. "Exposed! A Survey of Attacks on Private Data." *Annual Review of Statistics and Its Application* 4 (2017).

[94] Aaron Roth, Jonathan Ullman, and Zhiwei Steven Wu. "Watch and Learn: Optimizing from Revealed Preferences Feedback." *ACM SIGecom Exchanges* 14.1 (2015).

[95] Jonathan Ullman. "Query Release via Online Learning." *Encyclopedia of Algorithms* (2015).

Funding

Total Ullman share of funded research awards: approximately $2.3M.

Funded Grants and Awards

NSF Award #2247484 10/2023–09/2027

Project Title: *Auditing Private Statistical and Machine Learning Algorithms: Theory and Practice*
Role: Co-PI
PI: Alina Oprea (Northeastern)
Amount: $600K total; $300K Ullman share

NSF Award #2232692 04/2023–03/2028

Project Title: *Private Model Personalization*
Role: Lead PI
Co-PIs: Adam Smith (BU), Zhiwei Steven Wu (CMU)
Amount: $1.2M total; $450K Ullman share

NSF Award #2120603 10/2021–09/2023

Project Title: *Foundations for the Next Generation of Private Learning Systems*
Role: PI
Co-PIs: Alina Oprea (Northeastern), Adam Smith (BU), Roxana Geambasu (Columbia), Zhiwei Steven Wu (CMU)
Amount: $500K total; $100K Ullman share
Note: Funded for one year of the proposed five-year period.

Seed Grant from Apple, Inc. 04/2021

Project Title: *Auditing Differentially Private Machine Learning*
Role: PI
Co-PI: Alina Oprea (Northeastern)
Amount: $100K total; $50K Ullman share

NSF Award #1916020 10/2019–09/2022

Project Title: *Understanding and Mitigating the Privacy and Societal Risks of Advanced Advertising Targeting and Tracking*
Role: Co-PI
PI: Alan Mislove (Northeastern)
Co-PI: Aleksandra Korolova (USC)
Amount: $1.2M total; $400K Ullman share

NSF Award #1816028 09/2018–08/2022

Project Title: *New Approaches to Decentralized Differential Privacy*
Role: PI
Co-PI: Abhi Shelat (Northeastern)
Amount: $500K total; $250K Ullman share

NSF CAREER Award #1750640 02/2018–01/2024

Project Title: *CAREER: A Stable Foundation for Trustworthy Data Analysis*
Role: PI
Amount: $500K

Google Faculty Research Award 02/2018

Project Title: *Distributed Differential Privacy Beyond Local Protocols*
Role: PI
Amount: $65K

NSF Award #1718088 08/2017–07/2020

Project Title: *Programming Tools for Adaptive Data Analysis*
Role: PI
Co-PI: Marco Gaboardi (SUNY Buffalo)
Amount: $448K total; $224K Ullman share

Mentoring

Postdoctoral Researchers

Moshe Shenfeld 09/2026–Current

Joint with Adam Smith (Boston University)

Mahdi Haghifam 09/2023–08/2025
Fellowship: Khoury Distinguished Postdoctoral Fellow
Next Position: Research Assistant Professor, Toyota Technological Institute at Chicago

Maryam Aliakbarpour 08/2021–08/2023
Joint with Adam Smith (Boston University)
Next Position: Assistant Professor, Rice University

Audra McMillan 06/2018–06/2020
Fellowship: Cybersecurity & Privacy Institute Fellow
Joint Appointment: Boston University
Next Position: Research Scientist, Apple, Inc.

Doctoral Students

Youran Ye 09/2026–Current
Co-advisor: Lunjia Hu

Anagha Gokul 09/2026–Current
Co-advisor: Lunjia Hu

Rose Silver 09/2021–Current
Co-advisor: Elaine Shi (CMU)
Institution: CMU (transferred from Northeastern in 2023)

John Abascal 09/2021–08/2026
Thesis: *Quantifying Privacy Risks When Sharing Knowledge in Machine Learning*
Co-advisor: Alina Oprea
Next Position: Research Engineer, IBM Inc.

Konstantina Bairaktari 09/2020–12/2025
Thesis: *Theoretical Foundations of Multitask Learning and Metalearning*
Co-advisor: Huy Lê Nguyễn
Next Position: Postdoctoral Researcher, Aarhus University

Lydia Zakynthinou 09/2017–08/2023
Thesis: *Algorithms and Frameworks for Preventing Privacy Leakage and Overfitting in ML*
Co-advisor: Huy Lê Nguyễn
Awards: Facebook Fellowship; Khoury Graduate Research Award
Next Position: Postdoctoral Researcher, UC Berkeley
Current Position: Assistant Professor, Johns Hopkins University

Albert Cheu 09/2016–04/2021
Thesis: *Differential Privacy in the Shuffle Model*
Awards: Khoury Graduate Researcher Award
Next Position: Postdoctoral Researcher, Georgetown University
Current Position: Research Scientist, Google Inc.

Vikrant Singhal 09/2016–07/2021
Thesis: *Differentially Private Statistical Estimation*
Next Position: Postdoctoral Researcher, University of Waterloo
Current Position: Assistant Professor, University of Copenhagen

Undergraduate Researchers

Nicolás Berrios 06/2024–08/2024
Role: Undergraduate student, Pontifical Catholic University of Chile
Paper: *Black-Box Privacy Attacks on Shared Representations in Multitask Learning* in *ICLR 2026*
Next Position: Ph.D. in Computer Science, Georgia Tech

Eric Chapdelaine 06/2023–04/2024
Role: Undergraduate honors thesis student, Northeastern
Next Position: Software Engineer at Wayfair

Stanley Wu 01/2021–08/2023
Role: Undergraduate research assistant, Northeastern
Co-advisor: Alina Oprea
Paper: *How to Combine Membership-Inference Attacks on Multiple Updated Machine Learning Models* in *PETS 2023*
Next Position: Ph.D. in Computer Science, University of Chicago

Tatiana Ediger 01/2021–06/2021
Role: Undergraduate research assistant, Northeastern
Next Position: Data Engineer at Eight Sleep

Jeffrey Champion 06/2017–12/2019
Role: Undergraduate co-op student, Northeastern
Paper: *Securely Sampling Biased Coins with Applications to Differential Privacy* in *CCS 2019*
Next Position: Ph.D. in Computer Science, University of Texas at Austin

Mitali Bafna 06/2016–12/2016
Role: Undergraduate student, IIT Madras
Paper: *The Price of Selection in Differential Privacy* in *COLT 2017*
Next Position: Ph.D. in Computer Science, Harvard University

Thesis Committees

- Hongjie Chen (ETH Zurich; advisor: David Steurer) Summer 2026
- Harsh Chaudhari (advisor: Alina Oprea) Summer 2026
- Ethan Mook (advisor: Daniel Wichs) Spring 2026
- Lisa Oakley (advisor: Alina Oprea) Summer 2025
- Liudas Panavas (advisor: Cody Dunne) Spring 2025
- Schuyler Rosefield (advisor: Abhi Shelat) Fall 2024
- Matthew Jones (advisor: Huy Lê Nguyễn) Fall 2024
- Eysa Lee (advisor: Abhi Shelat) Summer 2023
- Jack Doerner (advisor: Abhi Shelat) Fall 2022
- Yashvanth Kondi (advisor: Abhi Shelat) Summer 2022
- Matthew Jagielski (advisors: Cristina Nita-Rotaru and Alina Oprea) Summer 2021
- Ariel Hamlin (advisor: Daniel Wichs) Spring 2021
- Giorgos Zirdelis (advisor: Daniel Wichs) Spring 2021
- Cheng Li (advisor: Jay Aslam) Fall 2019
- Adam Sealfon (MIT; advisor: Shafi Goldwasser) Summer 2019
- Chin Ho Lee (advisor: Emanuele Viola) Summer 2019
- Maryam Aziz (advisor: Jay Aslam) Fall 2018
- Benjamin Kreuter (advisor: Abhi Shelat) Spring 2018
- Zahra Jafargholi (advisor: Daniel Wichs) Spring 2016

Teaching

All courses taught at Northeastern University.

CS 3000: Algorithms & Data

Undergraduate course in algorithm design and analysis.
Formerly: CS 4800 Algorithms & Data

Term	Enrollment	Rating (Out of 5.0)
Spring 2026	78	4.2
Fall 2024	117	4.0
Fall 2023	124	4.2
Fall 2020 (4 Instructor Team)	321	4.6
Spring 2020	70	4.8
Fall 2018 (Sec 01)	72	4.7
Fall 2018 (Sec 04)	38	4.5
Spring 2018	48	4.8

CS 7800: Advanced Algorithms

Ph.D.-level core course in algorithm design and analysis.

Term	Enrollment	Rating (Out of 5.0)
Fall 2025	10	4.8
Fall 2022	28	4.8
Fall 2017	27	4.5
Fall 2016	28	4.9
Fall 2015	17	4.5

CS 7870: Seminar on Trustworthy Machine Learning

Ph.D.-level seminar in trustworthy machine learning.

Term	Enrollment	Rating (Out of 5.0)
Spring 2025	13	N/A

CS 7880: Differential Privacy in Machine Learning and Statistics

Ph.D.-level course in differential privacy.
Formerly: CS 7880 Rigorous Approaches to Data Privacy

Term	Enrollment	Rating (Out of 5.0)
Spring 2021	15	5.0
Spring 2017	9	5.0

Talks Given

Invited Talks

- European Workshop on Theory of Differential Privacy — Invited Plenary Talk 09/2026
- New England Statistics Symposium 05/2026
- Simons Institute Workshop on Trust in Decentralized Systems 03/2026
- University of Michigan CSE Seminar 11/2024
- UT Austin CS Colloquium 11/2024
- Boston University CDS Colloquium 03/2024
- University of Toronto CS — Distinguished Lecture Series 02/2024
- University of Wisconsin CS Seminar 01/2024

- TTIC Colloquium 07/2023
- Workshop on Privacy Preserving AI 02/2023
- UT Austin Computer Science Colloquium 03/2022
- University of Pennsylvania CIS Colloquium 02/2022
- TTIC Colloquium 10/2021
- Charles River Crypto Day 07/2021
- Apple Privacy-Preserving Machine Learning Workshop 08/2020
- IMA Workshop on Recent Themes in Resource Tradeoffs 06/2019
- University of Pennsylvania Warren Center Seminar 11/2018
- Banff Mathematical Foundations of Differential Privacy Workshop 05/2018
- Simons Statistics, Optimization, and Uncertainty Workshop 12/2017
- Simons Differential Privacy Semester Planning Workshop 05/2017
- IAS Symposium on Four Facets of Differential Privacy 11/2016
- PCMI Summer Session, the Mathematics of Data 07/2016
- IHP Nexus of Information and Computation Theories 03/2016
- NIPS Workshop on Adaptive Data Analysis 12/2015
- Oberwolfach Meeting on Complexity — Invited Plenary Talk 11/2015
- Theory and Practice of Differential Privacy Workshop — Invited Keynote Talk 04/2015
- Charles River Privacy Day, Boston University 05/2014
- Northeastern University CS Colloquium 03/2014
- University of Toronto CS Colloquium 02/2014
- Simons Big Data and Differential Privacy Workshop 03/2013
- Northwestern CS Colloquium 02/2013
- USC CS Colloquium 02/2013
- Simons Workshop on the Science of Differential Privacy 02/2013
- China Theory Week 10/2011

Seminar Talks

- University of Waterloo Theory Seminar 10/2025
- Tufts Theory Group Seminar 05/2025
- UMass Amherst Security Seminar 09/2024
- NTT Labs Seminar 07/2022
- University of Washington Computer Science Seminar 11/2021
- Google Differential Privacy Seminar 06/2021
- Boston University Theory of Computing Seminar 10/2019
- MIT Cryptographic and Information Security Seminar 04/2017
- UMass Amherst Database Seminar 10/2016
- MIT Theory of Computing Seminar 10/2016
- University of Warwick Theory of Computing Seminar 04/2015
- NYU Polytech Theory of Computing Seminar 10/2014
- Penn State Theory of Computing Seminar 09/2014
- MSR SVC Seminar 11/2013
- Princeton Theory of Computing Seminar 11/2013
- NYU Theory of Computing Seminar 10/2013
- Columbia University Theory of Computing Seminar 10/2013
- Boston University Theory of Computing Seminar 02/2011
- MSR New England Seminar 11/2010
- Penn State Theory of Computing Seminar 04/2010

Professional Activities

External Activities

Organizing

- Steering Committee Member, Association for Algorithmic Learning Theory 2026–Current
- Founder and Contributor, DifferentialPrivacy.org 2020–Current
- Steering Committee Co-chair, Annual Workshop on the Theory and Practice of Differential Privacy (TPDP) 2017–Current
- Co-organizer, Boston Differential Privacy Summer School 2022
- Instructor, 7th Bar Ilan University Winter School on Cryptography 2017

Program Committee Chair and Track Chair

- Program Committee Co-chair, Algorithmic Learning Theory 2026
- Track Chair, Conference on Computer and Communications Security (CCS), Privacy Track 2021
- Program Committee Chair, Workshop on Theory and Practice of Differential Privacy 2017

Program Committee and Area Chair

- Senior Area Chair, International Conference on Machine Learning (ICML) 2027
- Program Committee, Foundations of Responsible Computing (FORC) 2027
- Senior Program Committee, Algorithmic Learning Theory (ALT) 2027
- Program Committee, Secure and Trustworthy Machine Learning (SaTML) 2027
- Senior Area Chair, International Conference on Machine Learning (ICML) 2026
- Program Committee, Foundations of Responsible Computing (FORC) 2026
- Senior Area Chair, International Conference on Machine Learning (ICML) 2025
- Senior Program Committee, Conference on Learning Theory (COLT) 2025
- Program Committee, Symposium on Theory of Computing (STOC) 2025
- Area Chair, Conference on Learning Theory (COLT) 2024
- Program Committee, Foundations of Responsible Computing (FORC) 2024
- Program Committee, Symposium on Discrete Algorithms (SODA) 2024
- Area Chair, Conference on Learning Theory (COLT) 2023
- Area Chair, Conference on Fairness, Accountability, and Transparency (FAccT) 2022
- Senior Meta-Reviewer, Conference on Artificial Intelligence (AAAI) 2022
- Program Committee, Symposium on Security and Privacy (Oakland) 2021
- Reviewer, Conference on Learning Theory (COLT) 2020
- Program Committee, Conference on Computer and Communications Security (CCS) 2020
- Reviewer, International Conference on Machine Learning (ICML) 2020
- Program Committee, Information Theoretic Cryptography (ITC) 2020
- Program Committee, Symposium on Foundations of Computer Science (FOCS) 2018
- Reviewer, International Conference on Machine Learning (ICML) 2018
- Program Committee, Symposium on Discrete Algorithms (SODA) 2018
- Program Committee, Conference on Artificial Intelligence and Statistics (AISTATS) 2017
- Reviewer, Neural Information Processing Systems (NeurIPS) 2017
- Reviewer, Neural Information Processing Systems (NeurIPS) 2016
- Program Committee, Theory of Cryptography Conference (TCC) 2016-B
- Program Committee, Economics and Computation (EC) 2016
- Program Committee, Innovations in Theoretical Computer Science (ITCS) 2015
- Program Committee, Theory of Cryptography Conference (TCC) 2015
- Program Committee, Symposium on Theory of Computing (STOC) 2015

Editorial

- Editorial Board, Journal of Privacy and Confidentiality (JPC) 2025–Current
- Action Editor, Transactions on Machine Learning Research (TMLR) 2023–Current
- Guest Editor, *Journal of Privacy and Confidentiality* 2017

Advisory

- Member, Census Scientific Advisory Committee (CSAC) 2025

Internal Activities

Area Chair for Theory Research 09/2022–08/2026

Ex-officio member of the TT Hiring Committee

Khoury College of Computer Sciences Committees

- Representative, TT Hiring Committee 2022–2023 AY
- Chair, Teaching Awards Committee 2021–2022 AY
- Representative, Ph.D. Admissions Committee 2020–2021 AY
- Representative, Ph.D. Curriculum Committee 2020–2021 AY
- Representative, Teaching Awards Committee 2020–2021 AY
- Co-organizer, Ph.D. Open House 2019–2020 AY
- Representative, Ph.D. Admissions Committee 2019–2020 AY
- Representative, Ph.D. Curriculum Committee 2018–2019 AY
- Representative, TT Hiring Committee 2017–2018 AY
- Representative, TT Hiring Committee 2016–2017 AY
- Co-organizer, Ph.D. Open House 2015–2016 AY
- Representative, Ph.D. Admissions Committee 2015–2016 AY